

September 8th Colloquium Notes (ASTR457 Submission)

- Gaia's resolution is incredibly high, can map out bright areas across the sky and have a full map every month
 - Gaia has since retired, so there's no more data collection, just upcoming data releases
- 5000 candidates of star orbiting around white dwarf, 50 candidates of a star orbiting around neutron star, and 5 candidates of a star orbiting around black hole
- Tracked Gaia BH1 - sunlike star orbiting companion that is 10x bigger
 - What happens when that star becomes a red supergiant? Triggers a common envelope solution
 - Other ideas: What if the star won't expand to become a red giant or encounter companions? Did the binary star start as a triple? Did the black hole form with a single star?
- Tracked Gaia BH2: Period of about 3.5 years, 5AU orbit
- Tracked Gaia BH3: Metallicity is -2.5, but very large at about 33 solar masses
 - Part of a stellar stream
 - Could have formed dynamically or have had another companion
 - Stars in same cluster will have similar metallicity
 - BH1 and BH2 are pretty similar, BH3 is very different
- Drew a line - Linear Regression
 - **Question 1: How can you draw a line between data from only 3 points? This seems stretched significantly**
 - Start at 100 solar masses, when you collapse, almost all mass is lost in solar winds
- Testing showed that its not an interlockhold binary, like noise on a keplerian orbit
 - Why are the distances of a black hole from us and black hole period related? Did I misunderstand?
- Lithium enhancement in neutron star companionships
 - **Question 2: Is all this lithium being generated by the neutron star companionships, and how much more lithium are we seeing than what we would expect? Is the lithium able to come or leave from this system?**
- Black hole masses range between 1.5-2.5 (NS), ~8 (Gaia BH1, 2) and greater (BH3)
 - Neutron stars are not all born with the same mass
- Hubble Space Telescope: low metallicity star, does not fall on the line drawn for the Gaia black holes
- Black holes need different strengths of kicks, some very weak
 - B44 Signy has a companion!
 - Different modes of black hole formation
- Coming next: DR4 and Roman
 - Could have twice as many data points to search through , Will have much more detectability for black holes
 - DR3 could not find nearly as many
 - Roman: Will produce light curves of the galactic bulge & collapse uncertainty